

Hein Zarni Aung

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EDUCATION

Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Systems Fundamentals I & II, Programming Abstractions, Algorithms

INTERNSHIPS

Software Engineering Intern — Clustr

Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows, like/dislike interactions, and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation (Suggestions, Requests, Responses), a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.
- Implemented secure authentication features, including a password reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors to schedule and host lessons in one platform, and built a seed-data workflow to streamline debugging and local development.

PROJECTS

TuneAcademy (React, NumPy, Essentia, Firestore)

- Built the AI/audio-analysis pipeline for an app matching music students with instructors, extracting pitch, rhythm, and tone features with Essentia/NumPy, then pivoting from a planned ML model to rule-based scoring under hackathon time and compute constraints, delivering reliable detection within the 36-hour build window.
- Set up a CI/CD pipeline using GitHub Actions to lint and test the codebase, then auto-build and deploy the app to Cloudflare Workers.

Discord Voice Channel Activity Monitor (Node.js, Discord.js, Cloud, PM2, AWS)

- Built a Node.js/Discord.js bot to solve a scheduling problem for a friend group spread across different schedules, notifying users in real time when someone joins a voice channel, with customizable DM notifications, VIP lists, and per-server/per-user channel thresholds via JSON-based configuration.
- Deployed on a cloud server using PM2; reached 50 concurrent active users.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University

Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations; created original review materials adopted for subsequent semesters and led midterm review sessions reaching 100+ students combined in-person and via Zoom.

SKILLS

Languages: C, Python, JavaScript, TypeScript, Java

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: MongoDB, PostgreSQL, Firebase/Firestore, AWS, GCP

Tools: Git, Linux, Docker, PM2, Selenium